About

P19 - BI mini-project is an individual project issues as part of the BI course at VLBA, UOL.

Submission, evaluation and milestones

Please, refer to the information in the "P00 - Requirements (BI mini-project).pdf"

Data

Download your dataset on **financial-transactions** using the following link:

- https://cloud.uol.de/s/DRBP72f7odnrN6k

Tasks

- Review the given dataset
  - Make sure, that you can access the data
  - Check README.md
  - Study given dataset
  - Make sure, that you understand the value of the data and process behind the data
  - **NOTE**: *The dataset must be split into multiple tables*
- Create a **star schema** model for the given dataset
  - identify facts within the dataset (could be multiple)
  - identify respected dimensions
  - make sure that the schema reflects the data and no data get lost
- Create a **database schema**, that is based on the designed schema model
- ETL
  - Extract all relevant data fields from the given dataset
  - Transform extracted all relevant data
  - Import transformed data into created database
- Visualization/KPI
  - design and calculate KPI to insure the import of the data was successful
  - perform tasks (see the section below)
  - calculate KPI (see the section below)

Tasks

- What is the count of unique transactions per year, month, week?
- Which Merchant Category Code (MCC) generates the highest total transaction amount, and what is the corresponding MCC description?
- Calculate the average transaction amount for the `Utilities - Electric, Gas, Water, Sanitary` (MCC - 4900).
- What is the average yearly income (yearly_income) for all clients?
- What is the average yearly income (yearly_income) for clients, who have been issued more than two cards (num_cards_issued)?

- Identify the top 10 clients with the highest total debt (total_debt) and retrieve their corresponding average credit score.
- Calculate the average credit limit for clients whose card is flagged as being on the dark web (card_on_dark_web = 'Yes') versus those whose cards are not.
- Determine the average per capita income (`per_capita_income`) for customers grouped by their gender.
- What is the average total debt (total_debt) for clients whose `current_age` is within 10 years of their `retirement_age`?
- Calculate the percentage of transactions that were refunds (transaction amount is negative) for the department stores (MCC - 5311) .
- Create one task or business question of your own, and then answer it.

KPI

- What is the total transaction volume (sum of amount) across all transactions in the dataset?
- What is the average credit score of all clients in the dataset?
  What is the distribution of card brands (`card_brand`), expressed as a count of issued cards for each brand (e.g., Visa, Mastercard)?
- What is the average transaction amount (excluding refunds/negative values) for the entire dataset?
- Create one custom KPI on your own. Describe how to calculate it, and add it to the dashboard.